BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Chemistry — SSC-I (9th Grade)

Syllabus: Chapter 9 — Chemical Equilibrium

ANSWER KEY (Version 1)

Total Marks: 65	**Time Allowed: 2h 30m**	**Version: 1**

SECTION A — Multiple Choice Questions Answer Key (12 × 1 = 12 Marks)

Q1. Which is true about the equilibrium state? √ **D — Both reactions continue at the same rate** **1 mark**

The textbook defines chemical equilibrium as "**the state of a chemical reaction where the forward and reverse reactions take place at the same rate.**" The reactions do not stop (Key Points, p. 138). Options A, B, and C all imply reactions stopping, which is incorrect. Equilibrium is *dynamic*, not static.

Q2. A reversible reaction is represented by the symbol: √ **C — ⇄ (double arrow)** **1 mark**

The textbook states: "**A double arrow ⇄ in a chemical equation indicates that the reaction is reversible.**" (p. 135). A single arrow (→) denotes an irreversible/complete reaction. Options B and D show variations that are not the standard double-arrow symbol used in the textbook.

Q3. In an irreversible reaction, equilibrium is: √ **C — never established** **1 mark**

This is directly from Review Question 1(v) (p. 139). In an irreversible (complete) reaction, reactants are fully converted to products with no back-reaction possible, so equilibrium between forward and reverse reactions is **never established**. Option D is a distractor: the reaction can stop without establishing equilibrium.

Q4. Which does NOT happen when a system is at equilibrium? √ **A — Forward and reverse reactions stop** **1 mark**

This is directly from Review Question 1(iv) (p. 139). At equilibrium: rates become equal (B is true), concentrations stop changing (C is true), and reaction continues in both directions (D is true). The only false statement is A: reactions do **not** stop at equilibrium.

Q5. Hydrated copper(II) sulphate $CuSO_4 \cdot 5H_2O$ is: √ **B — a blue solid** **1 mark**

Key Points (p. 138): "**Hydrated copper(II) sulphate is a blue solid.**" Anhydrous $CuSO_4$ is white. The pink solid belongs to hydrated cobalt(II) chloride, which is a common distractor. This colour difference is testable in Activities.

Q6. When $CuSO_4 \cdot 5H_2O$ is heated, the product formed is: √ **A — $CuSO_4$ (white solid)** **1 mark**

The textbook activity states: "When copper(II) sulphate is heated, water is removed, forming **anhydrous copper(II) sulphate, a white solid**." (p. 137). Options B and C are incorrect: heat does not decompose the sulphate to CuO or Cu at these temperatures.

Q7. Anhydrous cobalt(II) chloride $CoCl_2$ is: √ **B — a blue solid** **1 mark**

Key Points (p. 138): "**Anhydrous cobalt(II) chloride is a blue solid.**" Hydrated $CoCl_2 \cdot 6H_2O$ is pink. Students frequently confuse the hydrated and anhydrous colours of copper sulphate vs. cobalt chloride: for cobalt, anhydrous = blue and hydrated = pink; for copper sulphate, it is the reverse (anhydrous = white, hydrated = blue).

Q8. When water is added to anhydrous $CoCl_2$, the colour change is: √ **B — blue to pink** **1 mark**

The textbook states: "When water is added to anhydrous cobalt(II) chloride (blue), it absorbs water and equilibrium shifts to the left to form hydrated $CoCl_2 \cdot 6H_2O$ (**pink**)." (p. 137). So the colour change is **blue (anhydrous) to pink (hydrated)**. The reverse (pink to blue) occurs on heating.

Q9. $H_2 + I_2 \rightleftarrows 2HI$ at equilibrium: substances present are: √ **D — H_2, I_2, and HI** **1 mark**

This is from Review Question 1(ii) (p. 138). At equilibrium, **both reactants and products are always present** because the forward and reverse reactions continue simultaneously. H_2 and I_2 have not fully reacted (the forward reaction is not complete), and HI is decomposing back. Option A would only be correct before the reaction starts.

Q10. Concentrations at equilibrium remain unchanged provided: √ **D — all three conditions are maintained** **1 mark**

This is from Review Question 1(iii) (p. 138). Section 9.2 lists three conditions that must all be met simultaneously: (1) concentration of reactants or products remains unchanged, (2) temperature is constant, and (3) pressure/volume is constant. Changing any one of these disrupts equilibrium.

Q11. When $CoCl_2 \cdot 6H_2O$ is heated, equilibrium shifts right because: √ **B — Heating favours the endothermic (forward) reaction to minimise temperature increase** **1 mark**

The textbook states: "When temperature is increased, equilibrium will shift to favour reactions which will **decrease the temperature**. So, **endothermic reaction is favoured** to minimise the change." (p. 137). Dehydration of $CoCl_2 \cdot 6H_2O$ (forward direction) is endothermic, so it is favoured on heating. This is Le Chatelier's principle applied to heat.

Q12. Chemical equilibrium is called *dynamic* because: √ **C — reactions do not stop; molecules are constantly reacting but concentrations remain constant** **1 mark**

The textbook: "**Chemical equilibrium is dynamic equilibrium. This is because reactions do not stop when they reach equilibrium. Individual molecules are constantly reacting.** However, the actual quantities of reactants and products do not change." (p. 136). Concentrations changing (A) would mean equilibrium is not yet reached. Only the forward reaction occurring (B) would describe an irreversible reaction.

SECTION B — Short Questions Answer Key (11 × 3 = 33 Marks)

2(i) — Reversible reaction: definition + two examples **3 marks**

Definition: A reaction in which **the products can react together to re-form the original reactants** is called a reversible reaction. A reversible reaction proceeds in both the forward and reverse directions under the same conditions. **1 mark**

Symbol: A double arrow (⇄) is used to indicate that the reaction is reversible. **1 mark**

Two examples (any two from the textbook): **1 mark**

```
(i) 2SO2(g) + O2(g) ⇌ 2SO3(g) (V2O5, 450°C, 200 atm) (ii) N2(g) + 3H2(g) ⇌ 2NH3(g) (Fe, 400°C, 200 atm) (iii)N2(g) + 2O2(g) ⇌ 2NO2(g) (electric spark)
```

OR

Complete reaction: A reaction in which **all reactants are converted completely into products** and the reaction does not reverse. The reaction goes to completion. **1 mark**

Difference: In a complete reaction, products cannot re-form reactants; in a reversible reaction, products react to give back original reactants simultaneously. A single arrow (→) shows a complete reaction; a double arrow (⇄) shows a reversible reaction. **1 mark**

Example of complete reaction: $2H_2 + O_2 \rightarrow 2H_2O$ (combustion). **Example of reversible reaction:** $N_{2(g)} + 3H_{2(g)} \rightleftarrows 2NH_{3(g)}$. **1 mark**

2(ii) — Chemical equilibrium defined; dynamic nature; two characteristics **3 marks**

Definition: "The state of a chemical reaction where the forward and reverse reactions take place at the same rate is called chemical equilibrium." (textbook p. 136) **1 mark**

Why dynamic: Equilibrium is dynamic because the reactions do *not* stop at equilibrium. Individual molecules are constantly reacting in both directions, but because the rate of the forward reaction equals the rate of the reverse reaction, the concentrations of all species remain constant. **1 mark**

Two characteristics of the equilibrium state: **1 mark**

1. The concentrations of reactants and products become constant (do not change over time).
2. Both forward and reverse reactions continue to occur simultaneously at the same rate.

OR

Using $2SO_{2(g)} + O_{2(g)} \rightleftarrows 2SO_{3(g)}$:

Initial state: Only SO_2 and O_2 are present. The rate of the forward reaction (forming SO_3) is at its highest because reactant concentration is maximum. The rate of the reverse reaction is zero (no SO_3 present yet). **1 mark**

As reaction progresses: SO_3 begins to form. The concentration of SO_2 and O_2 decreases (forward rate slows). SO_3 concentration increases and it begins to decompose back (reverse rate increases). **1 mark**

Equilibrium: Eventually the rate of the forward reaction equals the rate of the reverse reaction. Concentrations of SO_2, O_2, and SO_3 all remain constant. Equilibrium is established. **1 mark**

2(iii) — Three conditions for equilibrium **3 marks**

From Section 9.2 of the textbook, the three conditions that must be maintained for equilibrium to persist are:

1. The **concentration** of any reactant or product remains unchanged. **1 mark**
2. The **temperature** of the system remains constant. **1 mark**
3. The **pressure or volume** of the system remains constant. **1 mark**

OR

Closed system: A closed system is one in which **no matter can enter or leave** but energy can be exchanged with the surroundings. **1 mark**

Why equilibrium requires a closed system: If the container were open, products (especially gases) would escape, preventing the reverse reaction from occurring. Without a reverse reaction, equilibrium can never be established. For example, in $N_{2(g)} + 3H_{2(g)} \rightleftarrows 2NH_{3(g)}$, if NH_3 escapes, no reverse reaction occurs. **1 mark**

Example: When SO_2 and O_2 gases are mixed in a **sealed container**, SO_3 forms and decomposes back; eventually equilibrium is established. If the container were open, SO_3 vapour would escape and equilibrium would not form. **1 mark**

2(iv) — Experiment with $CuSO_4 \cdot 5H_2O$: colour changes and equation **3 marks**

Colour changes:

1. Initial colour of $CuSO_4 \cdot 5H_2O$: **blue**. When heated, colour changes from **blue to white** as water is driven off to form anhydrous $CuSO_4$. **1 mark**
2. When a few drops of water are added to the white anhydrous solid, colour changes from **white back to blue**, confirming the reaction is reversible. **1 mark**

Equation:

```
CuSO4·5H2O ⇌ CuSO4 + 5H2O (blue solid) (white solid) Heating → ← Cooling (+ water)
```

1 mark

OR

Anhydrous $CuSO_4$ as a test for water: Anhydrous copper(II) sulphate is a **white solid**. When water is added, it absorbs the water and converts back to hydrated $CuSO_4 \cdot 5H_2O$, which is **blue**. **1 mark**

Confirmation: A colour change from **white to blue** confirms the presence of water. This is a simple, visual test for water. **1 mark**

Equation:

```
CuSO4 (white) + 5H2O → CuSO4·5H2O (blue)
```

1 mark

2(v) — Cobalt(II) chloride equilibrium: colours, heating effect **3 marks**

Colours (from Key Points, p. 138):

Hydrated $CoCl_2 \cdot 6H_2O$ is a **pink solid**. **1 mark**
Anhydrous $CoCl_2$ is a **blue solid**. **1 mark**

Effect of heating: When the pink hydrated cobalt(II) chloride is heated, it loses its water of crystallisation and becomes anhydrous $CoCl_2$ (blue). The equilibrium shifts towards the right (towards the anhydrous form). This is because increasing temperature favours the endothermic reaction (dehydration). **1 mark**

OR

Equation:

```
CoCl2·6H2O ⇌ CoCl2 + 6H2O (pink solid) (blue solid) Heating → ← Cooling (+ water)
```

1 mark

Adding water to anhydrous $CoCl_2$: When water is added to the **blue** anhydrous $CoCl_2$, it absorbs water and the equilibrium shifts to the **left** (reverse direction), forming hydrated $CoCl_2 \cdot 6H_2O$. The colour changes from **blue back to pink**. **1 mark**

Forward reaction: $CoCl_2 \cdot 6H_2O \rightarrow CoCl_2 + 6H_2O$ (dehydration, on heating).
Reverse reaction: $CoCl_2 + 6H_2O \rightarrow CoCl_2 \cdot 6H_2O$ (rehydration, on adding water). **1 mark**

2(vi) — Effect of increasing temperature on equilibrium **3 marks**

General principle (Section 9.3): "When temperature is increased, equilibrium will shift to favour the reactions which will **decrease the temperature**. So, endothermic reaction is favoured to minimise the change." **1 mark**

Direction of shift: The equilibrium shifts in the direction that **absorbs heat** (the endothermic direction), thereby opposing the temperature increase. **1 mark**

Example using $CuSO_4·5H_2O$: Heating $CuSO_4·5H_2O$ shifts the equilibrium to the right (forward direction): $CuSO_4·5H_2O \rightarrow CuSO_4 + 5H_2O$. The dehydration reaction is endothermic (it absorbs heat). The blue colour changes to white. **1 mark**

OR

Effect of cooling: When a chemical equilibrium system is cooled (temperature decreased), the system opposes the change by shifting in the **exothermic direction** (the direction that releases heat), thereby raising the temperature back towards the original. **1 mark**

Cobalt(II) chloride equilibrium on cooling: For $CoCl_2·6H_2O \rightleftarrows CoCl_2 + 6H_2O$, on cooling the equilibrium shifts to the **left** (reverse direction). $CoCl_2$ reabsorbs water to re-form $CoCl_2·6H_2O$. The colour changes from **blue back to pink**. This is the exothermic (reverse) direction. **2 marks**

2(vii) — $CO + 3H_2 \rightleftarrows CH_4 + H_2O$: forward and reverse reactions **3 marks**

The full reversible reaction (from textbook exercise Q3, p. 139):

```
CO(g) + 3H2(g) ⇌ CH4(g) + H2O(g)
```

Forward reaction: The reaction in which reactants convert to products (left to right): **1 mark**

```
CO(g) + 3H2(g) → CH4(g) + H2O(g)
```

Coal reacts with hot steam to form CO and H_2. These react with a catalyst to give methane (CH_4) and water vapour. **1 mark**

Reverse reaction: The reaction in which products convert back to reactants (right to left): **1 mark**

```
CH4(g) + H2O(g) → CO(g) + 3H2(g)
```

OR

Reversible equation for BrCl decomposition (Think Tank, p. 139):

```
2BrCl(g) ⇌ Br2(g) + Cl2(g)
```

1 mark

Forward reaction: $2BrCl_{(g)} \rightarrow Br_{2(g)} + Cl_{2(g)}$. Bromine chloride decomposes to form bromine and chlorine gases. **1 mark**

Conditions favouring reverse reaction: Decreasing temperature (cooling) would favour the exothermic reverse reaction ($Br_2 + Cl_2 \rightarrow 2BrCl$). Also, increasing pressure (or increasing concentration of Br_2 and Cl_2) would shift equilibrium left, favouring formation of BrCl. **1 mark**

2(viii) — $N_2 + 3H_2 \rightleftarrows 2NH_3$: forward, reverse, conditions 3 marks

(a) Forward reaction (reactants to products, left to right): 1 mark

```
N2(g) + 3H2(g) → 2NH3(g) (Nitrogen + Hydrogen → Ammonia)
```

(b) Reverse reaction (products back to reactants, right to left): 1 mark

```
2NH3(g) → N2(g) + 3H2(g) (Ammonia decomposes back to Nitrogen and Hydrogen)
```

Conditions from textbook (p. 135): Catalyst **Fe (iron)**, temperature **400°C**, pressure **200 atm**. 1 mark

OR

(a) Forward reaction for SO_2/SO_3 equilibrium: 1 mark

```
2SO2(g) + O2(g) → 2SO3(g) (Sulphur dioxide + Oxygen → Sulphur trioxide)
```

(b) Reverse reaction: 1 mark

```
2SO3(g) → 2SO2(g) + O2(g) (Sulphur trioxide decomposes back)
```

Industrial conditions (textbook p. 135): Catalyst **V_2O_5 (vanadium pentoxide)**, temperature **450°C**, pressure **200 atm**. This is the Contact Process for manufacturing sulphuric acid. 1 mark

2(ix) — Fizzy drinks CO_2 example: physical equilibrium 3 marks

From Science Titbits, p. 136:

Equilibrium equation:

```
CO2(g) ⇌ CO2(aq)
```

1 mark

When sealed: CO_2 is dissolved in the liquid drink under pressure. The forward reaction ($CO_{2(g)} \rightarrow CO_{2(aq)}$) and reverse reaction ($CO_{2(aq)} \rightarrow CO_{2(g)}$) occur at the same rate. Equilibrium is maintained. 1 mark

When lid is removed: The pressure above the liquid drops suddenly. The equilibrium shifts to the left (towards the gaseous phase). CO_2 dissolved in the liquid comes out of solution and **bubbles of CO_2 appear**. When the lid is replaced, pressure is restored and bubbles stop: equilibrium re-establishes. 1 mark

OR

Liquid-vapour equilibrium in a closed container:

Initial state: Liquid is placed in a closed container. Initially the rate of **evaporation** is high (liquid molecules escape into vapour phase) and the rate of **condensation** is zero (no vapour present). 1 mark

As time progresses: Vapour accumulates above the liquid. More vapour molecules collide with the liquid surface and condense. Rate of condensation increases while rate of evaporation decreases (as liquid molecules escape). 1 mark

Equilibrium: Eventually the **rate of evaporation equals the rate of condensation**. The amounts of liquid and vapour remain constant. Equilibrium equation:

```
Liquid ⇌ Vapours
```

1 mark

2(x) — Forward vs reverse reactions; $N_2 + 2O_2 \rightleftarrows 2NO_2$ example 3 marks

Forward reaction: The reaction in which **reactants convert into products** (proceeds from left to right). It is shown by a forward arrow. 1 mark

Reverse reaction: The reaction in which **products convert back into the original reactants** (proceeds from right to left). It occurs simultaneously with the forward reaction in a reversible system. 1 mark

Example from $N_2 + 2O_2 \rightleftarrows 2NO_2$:

```
Forward: N2(g) + 2O2(g) → 2NO2(g) (Nitrogen + Oxygen → Nitrogen dioxide) Reverse: 2NO2(g) → N2(g) + 2O2(g) (Nitrogen dioxide → Nitrogen + Oxygen)
```

1 mark

OR

Using $2NO_{2(g)} \rightleftarrows N_{2(g)} + 2O_{2(g)}$:

Converts reactants to products: The **forward reaction** (left to right) converts NO_2 to N_2 and O_2: $2NO_{2(g)} \rightarrow N_{2(g)} + 2O_{2(g)}$. 1 mark

Converts products back to reactants: The **reverse reaction** (right to left) converts N_2 and O_2 back to NO_2: $N_{2(g)} + 2O_{2(g)} \rightarrow 2NO_{2(g)}$. 1 mark

At equilibrium: The rate of the forward reaction equals the rate of the reverse reaction. Concentrations of NO_2, N_2, and O_2 all become constant. Both reactions continue simultaneously at the same rate. 1 mark

2(xi) — Industries using equilibrium / Key Points from Chapter 9 3 marks

From the textbook introduction (p. 135): "Equilibrium reactions are involved in a number of steps in the commercial production of many important chemicals such as **ammonia** and **sulfuric acid**."

Industries and reactions relying on equilibrium: 1 mark each, up to 3 marks

1. **Ammonia (Haber Process):** $N_2 + 3H_2 \rightleftarrows 2NH_3$ (Fe catalyst, 400°C, 200 atm)
2. **Sulphuric acid (Contact Process):** $2SO_2 + O_2 \rightleftarrows 2SO_3$ (V_2O_5, 450°C, 200 atm)
3. **Nitrogen dioxide production:** $N_2 + 2O_2 \rightleftarrows 2NO_2$ (electric spark)

Importance: Understanding equilibrium allows industry to select conditions (temperature, pressure, catalyst) that maximise yield of the desired product while minimising energy costs.

OR

Key Points from Chapter 9 (p. 138): 1 mark per point, any 4

1. A reaction in which the products can react together to re-form the original reactants is called a **reversible reaction**.
2. A reversible reaction is shown by the double arrow symbol (⇄).
3. **Anhydrous copper(II) sulphate** is a **white solid**.
4. **Hydrated copper(II) sulphate** is a **blue solid**.
5. **Anhydrous cobalt(II) chloride** is a **blue solid**.
6. **Hydrated cobalt(II) chloride** is a **pink solid**.
7. A state of a chemical reaction in which forward and reverse reactions take place at the same rate is called **chemical equilibrium**.

SECTION C — Long Questions Answer Key (4 × 5 = 20 Marks)

3(i) — Reversible reactions, equilibrium definition + three conditions **5 marks**

(a) Definitions and distinction: [3 marks]

Reversible reaction: A reaction in which the products can react together to re-form the original reactants. It proceeds in both the forward and reverse directions under the same conditions and **never ends**. Shown by $\rightleftharpoons$. **1 mark**

Chemical equilibrium: The state of a chemical reaction where the forward and reverse reactions take place at the **same rate**. Concentrations of reactants and products remain constant. **1 mark**

Feature	Complete (Irreversible) Reaction	Reversible Reaction
Direction	One direction only ($\rightarrow$)	Both directions ($\rightleftharpoons$)
Completion	All reactants convert to products	Mixture of reactants and products at equilibrium
Equilibrium	Never established	Established in a closed system
Example	$2H_2 + O_2 \rightarrow 2H_2O$	$N_2 + 3H_2 \rightleftharpoons 2NH_3$

1 mark

(b) Three conditions for equilibrium: [2 marks]

1. Concentration of any reactant or product remains unchanged. **⅔ mark**
2. Temperature of the system remains constant. **⅔ mark**
3. Pressure or volume of the system remains constant. **⅔ mark**

Effect if one changes: If any condition is altered, the equilibrium is disturbed and the system shifts in the direction that opposes the change (to minimise the disturbance). For example, increasing temperature shifts equilibrium towards the endothermic direction. **1 mark**

OR

(a) Equilibrium established step by step in $2SO_2 + O_2 \rightleftharpoons 2SO_3$: [3 marks]

Step 1 (Initial state): Only SO_2 and O_2 are present in the sealed container. Rate of forward reaction is maximum; rate of reverse reaction = 0. No SO_3 has formed yet. **1 mark**

Step 2 (Reaction progresses): SO_3 begins to form (forward reaction). Concentration of SO_2 and O_2 decreases, slowing the forward rate. SO_3 begins to decompose (reverse reaction starts). Reverse rate increases. **1 mark**

Step 3 (Equilibrium): Forward rate = reverse rate. Concentrations of SO_2, O_2, and SO_3 all become constant. Equilibrium is established. The reaction continues in both directions but with no net change in concentrations. **1 mark**

(b) Dynamic vs static equilibrium: [2 marks]

Dynamic equilibrium: Both forward and reverse reactions continue to occur; molecules are constantly reacting, but concentrations remain constant because both rates are equal. **1 mark**
Static equilibrium: No reactions are occurring; everything has stopped. Chemical equilibrium is NOT static.
Analogy: Imagine escalators in a busy mall: if exactly as many people step on at the bottom as step off at the top each second, the number of people on the escalator stays constant, even though everyone is moving. The system is dynamic. **1 mark**

3(ii) — $CuSO_4 \cdot 5H_2O$ experiment + equation with labels 5 marks

(a) Laboratory experiment on $CuSO_4 \cdot 5H_2O$: [3 marks]

Materials required (from textbook p. 137): Test tube, dropper, heating system, hydrated copper(II) sulphate. **½ mark**

Procedure:

1. Place 5 g of hydrated copper(II) sulphate (blue) in a test tube and heat slowly. **½ mark**
2. Observe the colour change from **blue to white** as water is driven off. **½ mark**
3. Allow the test tube and contents to cool to room temperature.
4. Add a few drops of water using a dropper. Observe the colour change from **white back to blue**. **½ mark**

Observations:

- On heating: colour changes **blue → white** (hydrated to anhydrous). Water is released as vapour.
- On adding water: colour changes **white → blue** (anhydrous to hydrated). Heat may be felt (exothermic).

1 mark

Conclusion: The reaction is **reversible**. Heat drives the forward reaction; adding water drives the reverse reaction. **½ mark**

(b) Reversible equation with labels: [2 marks]

```
Heating → CuSO4·5H2O ⇌ CuSO4 + 5H2O (blue solid) (white solid) ← Cooling / adding water
```

1 mark (equation correct) + **1 mark** (colours and direction labels correct)

OR

(a) Cobalt(II) chloride equilibrium: [3 marks]

Colours: $CoCl_2 \cdot 6H_2O$ (hydrated) is **pink**; $CoCl_2$ (anhydrous) is **blue**. **1 mark**

(i) Effect of heating: When the pink hydrated solid is heated, water is removed. The equilibrium $CoCl_2 \cdot 6H_2O \rightleftarrows CoCl_2 + 6H_2O$ shifts to the **right**. Colour changes from **pink to blue**. The forward (dehydration) reaction is endothermic. **1 mark**

(ii) Effect of adding water: When water is added to the blue anhydrous $CoCl_2$, it absorbs water and equilibrium shifts to the **left**. $CoCl_2 \cdot 6H_2O$ reforms. Colour changes from **blue back to pink**. **1 mark**

(b) Equation and direction of shift on heating: [2 marks]

```
Heating → CoCl2·6H2O ⇌ CoCl2 + 6H2O (pink solid) (blue solid) ← Cooling / adding water
```

On heating, equilibrium shifts **right** (endothermic direction). The textbook principle: "When temperature is increased, equilibrium shifts to favour reactions which decrease the temperature (endothermic reaction)." **2 marks**

3(iii) — Temperature effects on equilibrium + effect of adding water **5 marks**

(a) General principle and predictions: [3 marks]

General principle (Section 9.3): When temperature is increased, the equilibrium shifts to favour the **endothermic reaction** (the reaction that absorbs heat), thus opposing the temperature increase. **1 mark**

For $CoCl_2·6H_2O \rightleftarrows CoCl_2 + 6H_2O$:
The forward reaction (dehydration) is endothermic (requires heat).
The reverse reaction (hydration) is exothermic (releases heat).

(i) On heating: Equilibrium shifts to the **right** (forward direction). More $CoCl_2$ (blue) and H_2O are formed. Colour changes from **pink to blue**. **1 mark**

(ii) On cooling: Equilibrium shifts to the **left** (reverse direction). More $CoCl_2·6H_2O$ (pink) is formed. Colour changes from **blue back to pink**. The exothermic reverse reaction is favoured. **1 mark**

(b) Effect of adding water (increasing concentration): [2 marks]
When water is added to the cobalt(II) chloride equilibrium system, the concentration of H_2O (a product) increases. The system opposes this change by shifting the equilibrium to the **left** (reverse direction), consuming the excess water and forming more $CoCl_2·6H_2O$ (pink). **1 mark**
This demonstrates that changing the concentration of a product shifts equilibrium towards the reactant side. The colour change from **blue to pink** confirms the leftward shift. **1 mark**

OR

(a) $N_2 + 3H_2 \rightleftarrows 2NH_3$: forward, reverse, and conditions: [3 marks]

Forward reaction:

```
N2(g) + 3H2(g) → 2NH3(g) (Nitrogen reacts with hydrogen to form ammonia)
```

1 mark

Reverse reaction:

```
2NH3(g) → N2(g) + 3H2(g) (Ammonia decomposes back to nitrogen and hydrogen)
```

1 mark

Conditions: Catalyst: **Fe (iron)**; Temperature: **400°C**; Pressure: **200 atm**. These conditions are chosen industrially to obtain a reasonable yield of ammonia at an acceptable rate. **1 mark**

(b) Closed system and equilibrium: [2 marks]
Closed system: A system in which no matter enters or leaves (but energy exchange is possible). **1 mark**
Open system disrupts equilibrium: In an open system, gaseous products escape into the atmosphere. The reverse reaction cannot occur because products are no longer present. Equilibrium is never established. For example, if NH_3 escaped from the reaction vessel in the Haber Process, N_2 and H_2 would keep reacting forward without a reverse reaction. **1 mark**

3(iv) — Key Points summary + reversible vs irreversible comparison table 5 marks

(a) Key Points of Chapter 9 (from p. 138): [3 marks]

1. A reaction in which the products can react together to re-form the original reactants is called a **reversible reaction**. ½
2. A reversible reaction is shown by the symbol ⇄ (double arrow). ½
3. **Anhydrous copper(II) sulphate** is a **white solid**. ½
4. **Hydrated copper(II) sulphate** is a **blue solid**. ½
5. **Anhydrous cobalt(II) chloride** is a **blue solid**. ½
6. **Hydrated cobalt(II) chloride** is a **pink solid**. ½
7. A state of a chemical reaction in which forward and reverse reactions take place at the same rate is called **chemical equilibrium**.

1 mark (7th point)

(b) Comparison table: [2 marks]

Heading	Reversible Reaction	Irreversible (Complete) Reaction
(i) Direction	Both forward and reverse simultaneously	Forward direction only
(ii) Equilibrium	Established in a closed system	Never established
(iii) Example	$CuSO_4 \cdot 5H_2O \rightleftarrows CuSO_4 + 5H_2O$	$2H_2 + O_2 \rightarrow 2H_2O$ (combustion)
(iv) Symbol	⇄ (double arrow)	→ (single arrow)

2 marks

OR

(a) Liquid-vapour equilibrium (Liquid ⇄ Vapours): [3 marks]

Start: Liquid is placed in a **closed container**. Some liquid evaporates: faster molecules escape from the surface into the vapour phase. Rate of evaporation is at its highest; rate of condensation = 0 (no vapour yet). 1 mark

Progress: As more liquid evaporates, vapour density above the liquid increases. Vapour molecules begin to collide with the liquid surface and condense back. Rate of condensation increases; rate of evaporation decreases (fewer energetic molecules remaining). 1 mark

Equilibrium: Rate of evaporation = rate of condensation. The amount of liquid and the amount of vapour both remain constant. This is **physical equilibrium**. Even though individual molecules continue to evaporate and condense, the macroscopic amounts do not change. 1 mark

(b) CO_2 fizzy drinks and pressure effect: [2 marks]

Equilibrium equation:

```
CO2(g) ⇌ CO2(aq)
```

(i) Lid sealed: CO_2 is under high pressure inside the bottle. The equilibrium favours the dissolved state (right): $CO_2(aq)$ is abundant; only a little $CO_2(g)$ is present. No bubbles form. 1 mark

(ii) Lid removed: Pressure above the liquid drops to atmospheric pressure. The equilibrium shifts to the **left** (towards gaseous CO_2): dissolved CO_2 comes out of solution and **bubbles appear**. Replacing the lid restores pressure and stops bubbling (equilibrium re-establishes). 1 mark